Chapter 4 – Second Normal Form (2NF)

Chapter Objectives

After completing this chapter, students will understand:

- Why 1NF is not enough
- What is 2NF?
- Functional Dependency Revision
- Full Functional Dependency
- Partial Dependency
- Composite Primary Key
- Why Partial Dependency is bad
- How to convert a table into 2NF
- SQL Implementation
- Interview Questions
- Practice Questions

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1. Why Do We Need 2NF?

Suppose we already converted our database into **1NF**.

Our table now looks like this.

StudentID	CourseID	StudentName	CourseName
101	C1	Akshat	SQL
101	C2	Akshat	Python
102	C1	Rahul	SQL

Every cell has one value.

So...

Is this table perfect?

No.

There is still unnecessary repetition.

Student name appears multiple times.

Course name appears multiple times.

This leads to redundancy.

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2. Definition of 2NF

A table is in **Second Normal Form (2NF)** if:

1. It is already in **1NF**

2. Every **non-prime attribute** depends on the **entire primary key**.

Understanding the Definition

Students usually get confused by this definition.

Let's simplify it.

Suppose your Primary Key is

```
(StudentID, CourseID)
```

This means **both columns together** uniquely identify a row.

Now ask every non-key column:

> **Do you need BOTH StudentID and CourseID to determine your value?**

If the answer is **No**, then the table is **not in 2NF**.

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3. Composite Primary Key

Remember

A Composite Primary Key consists of multiple columns.

Example

```
(StudentID, CourseID)
```

Neither StudentID nor CourseID alone is unique.

Together they uniquely identify a record.

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4. Functional Dependency Revision

Suppose

```
StudentID → StudentName
```

If StudentID is known

StudentName is automatically known.

Similarly

```
CourseID → CourseName
```

Knowing CourseID tells us the CourseName.

---

5. Full Functional Dependency

Suppose

Primary Key

```
(StudentID, CourseID)
```

Marks

depend on

both StudentID

and

CourseID.

```
(StudentID, CourseID) → Marks
```

Need both.

This is called

Full Functional Dependency.

Example

Student 101

SQL

Marks = 95

Student 101

Python

Marks = 88

Can StudentID alone determine marks?

No.

Can CourseID alone determine marks?

No.

Need both.

This is Full Dependency.

---

6. Partial Dependency

Now consider

```
StudentID → StudentName
```

Do we need CourseID?

No.

StudentID alone tells us the student's name.

Therefore

StudentName depends only on

part of the Primary Key.

This is called

Partial Dependency.

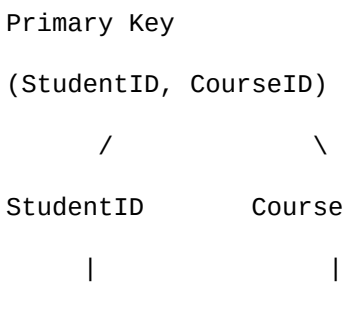
Easy Definition

If a non-key column depends on only part of a Composite Primary Key,

it is called

Partial Dependency.

Visual Diagram



```
StudentName     CourseName
```

Notice

StudentName depends only on StudentID.

CourseName depends only on CourseID.

Not on both.

This violates 2NF.

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7. Why Partial Dependency Is Bad

Suppose

Akshat changes his name.

Current table

StudentID	CourseID	StudentName
101	C1	Akshat
101	C2	Akshat

Need to update

both rows.

If one row is forgotten

StudentID	CourseID	StudentName
101	C1	Akshat
101	C2	Akshat Saini

Database becomes inconsistent.

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8. Converting to 2NF

Split the table.

Student

StudentID	StudentName
101	Akshat
102	Rahul

Primary Key

```
StudentID
```

Course

CourseID	CourseName
C1	SQL
C2	Python

Primary Key

```
CourseID
```

Enrollment

StudentID	CourseID
101	C1
101	C2
102	C1

Primary Key

```
(StudentID, CourseID)
```

Now

StudentName depends on StudentID.

CourseName depends on CourseID.

Enrollment stores only the relationship.

No partial dependency.

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9. SQL Implementation

Student

```
CREATE TABLE Student(
    StudentID INT PRIMARY KEY,
    StudentName VARCHAR(50)
);
```

Course

```
CREATE TABLE Course(
    CourseID VARCHAR(10) PRIMARY KEY,
    CourseName VARCHAR(50)
);
```

Enrollment

```
CREATE TABLE Enrollment(
    StudentID INT,
    CourseID VARCHAR(10),
    PRIMARY KEY(StudentID, CourseID),
    FOREIGN KEY(StudentID) REFERENCES Student(StudentID),
    FOREIGN KEY(CourseID) REFERENCES Course(CourseID)
);
```

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10. Important Interview Question

Can a table with a single-column Primary Key violate 2NF?

Answer

No.

Because

Partial Dependency is possible only when the Primary Key contains multiple columns.

Therefore

If a table has

- Single Primary Key
- Already in 1NF

It is automatically in 2NF.

Common Mistakes

✕ Thinking repeated values violate 2NF.

Wrong.

2NF only checks **dependencies**, not repeated values.

✕ Thinking every table must have a Composite Key.

Wrong.

Many tables have a single Primary Key.

✕ Thinking 2NF applies to every table.

Wrong.

2NF problems occur only when there is a Composite Primary Key.

Memory Trick

2NF asks only one question:

> **Does every non-key column depend on the WHOLE Composite Primary Key?**

If **Yes** → 2NF

If **No** → Not 2NF

Chapter Summary

- 2NF comes after 1NF.
- Removes Partial Dependency.
- Works mainly with Composite Keys.
- Every non-key attribute must depend on the whole Primary Key.